

Credit Card Fraud Detection Using Machine Learning

Model Development, Evaluation, and Ethical AI Assessment

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Executive Summary

This project developed a machine learning solution for detecting fraudulent credit card transactions using the ULB Credit Card Fraud dataset. The workflow included exploratory data analysis, preprocessing, feature selection, model comparison, hyperparameter tuning, and an Ethical AI assessment. XGBoost was selected as the final production candidate because it provided the best balance of predictive performance. Beyond model accuracy, the project evaluated explainability using SHAP, assessed potential bias across transaction groups, documented limitations, and proposed an engineering deployment strategy. The project concludes that Human-in-the-Loop validation, continuous monitoring, and periodic retraining are recommended before production deployment.

Project Overview

This project investigates the use of supervised machine learning for credit card fraud detection. The objectives were to identify an effective predictive model, understand the factors influencing its decisions, evaluate fairness and explainability, and determine whether the model is suitable for deployment in a financial environment.

Methodology

The workflow consisted of exploratory data analysis, preprocessing, feature selection using Mutual Information, Random Forest and SHAP, model development with Logistic Regression, Random Forest and XGBoost, hyperparameter tuning, model evaluation, and a final Ethical AI assessment.

Results and Discussion

Model comparison identified XGBoost as the strongest overall performer. SHAP analysis confirmed consistent feature importance patterns, while bias audits and fairness metrics

indicated generally consistent behaviour across observable transaction groups. These findings support confidence in the model but do not eliminate the need for further validation.

Ethical AI and Engineering Deployment

The project emphasized that predictive performance alone is insufficient for deployment. Explainability, fairness, transparency, and operational risk were evaluated. A phased deployment beginning with Human-in-the-Loop supervision, continuous monitoring, and periodic retraining is recommended.

Conclusion

This capstone demonstrated that machine learning can effectively support fraud detection while highlighting the importance of responsible AI. The selected model showed strong performance, but successful deployment requires explainability, fairness assessment, ongoing monitoring, and human oversight.

References

- Chen & Guestrin (2016) XGBoost.
- Lundberg & Lee (2017) SHAP.
- Kaggle ULB Credit Card Fraud Detection Dataset.
- Pedregosa et al. (2011) Scikit-learn.